

TrainAudit

OFFLINE — mine once per framework

C2 mine+verify

1 Invariant Miner



LLM-instantiated candidates, untrusted until verified



Instantiate

Scope

fix training phase
+ parameter class

Ground

evidence $\mathcal{E}^+/\mathcal{E}^-$ from
framework source & docs

verify (decisive)

Construct

counterexample attack
 ≥ 2 CEs; one CE holds
 $\Rightarrow$ reject outright

Accept

no CE holds $\wedge$
 $\text{Conf}(c) \geq 0.8$
($\theta = 0.8$)

reject → next iteration

five-stage FSM

S_1

S_2

S_3

S_4

S_5

gap analysis

evidence retrieval

synthesis+attack

persistence

reporting

✓ decisive counterexample gate

✓ healthy-run replay (verification only)

✓ $\text{Conf}(c) \geq 0.8$

ONLINE — every training job · fully deterministic · no LLM

C3 keep pace

2 Data Collector

no user code changes · adapter 30-150 LoC



+ auxiliary taps (ckpt · all_reduce · snapshot) = 8 hookpoints
GPU-side scalar reductions (checksums only when required)

rank / step / stage unrolled into columns
written to DuckDB trace (two-step window)

captured record (S0):

cksum

param_name

rank

step

stage

DuckDB trace DB

two-step sliding window

3 Verifier

deterministic SQL checking — no LLM

at job start

extract topology · topology-aware pruning · lineage: replicated vs. sharded

empty result = holds · non-empty IS the violation set

coarse → fine: checksum → module → param → bitwise

per step, per rule

WHERE $\#_{\text{topo}} \wedge \#_{\text{precond}}$
GROUP BY param, step
HAVING $\#_{\text{schema}}$ violated

non-empty violation set

param	rank	cksum
router.weight	0	cksum ₀
router.weight	1	cksum ₁

x divergent checksums

- ✓ violating constraint
- ✓ affected parameter
- ✓ divergent ranks
- ✓ first violating step



Verified Constraint Library $\mathcal{P}$

C1: topology + phase guards

P3 · cross-rank replication

(SwitchMLP router sync)

$\#_{\text{topo}}$: TP > 1
 $\#_{\text{precond}}$: tpl = False
(not tensor-parallel sharded)
 $\#_{\text{schema}}$: replicas share checksum

compiled to parameterized SQL at job start
accepted rules determine trace schema (S0-S6)
✓ 0 false positives over 764 clean evals

Pattern Catalog

35 relation templates
13 subsystem labels
induced from a corpus of
392 evidence records
versioned as new framework
code lands

acceptance gates

- ✓ counterexample attack
- ✓ healthy-run replay
- ✓ two independent sources
- ✓ $\text{Conf}(c) \geq 0.8$ screen

framework adapters

Megatron-LM · DeepSpeed
OLMo · OLMo-core · FSDP
30-150 LoC each; one pinned
library serves all five

Legend

offline mining

online checking

reject / violation

LLM only inside offline Miner · online path: no LLM

offline inputs

Pattern Catalog

template P3:
pi_topo: TP > 1
pi_precond: tpl=False
pi_schema: share cksum
(cross-rank replicas)
from 392 evidence records
35 templates · 13 subsystem labels

Framework source

Megatron-LM
DeepSpeed
OLMo / OLMo-core
silent-vs-loud filter
no raise / assert
once per framework

online output

Violation Report

violating constraint: P3
affected parameter:
router.weight
divergent ranks: 0 vs 1
(SwitchMLP router sync)
non-empty SQL result

empty result = holds
non-empty = violation set